Strategic Corporate Responses to Portfolio Reallocation: Evidence from Dividends

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October 30, 2026

Firms try to attract investors, but investor capital is sticky

Firms act to raise demand for their shares

- Accounting standards (Covrig et al., 2007)
- Investor relations (Bushee and Miller, 2012)
- Dividends (Baker and Wurgler, 2004)

Assumption: investor capital **responds**.

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If investor capital rarely moves, **how can firms influence it?**



Investors reconsider at **particular moments** → firms may **time** their actions.

This paper: dividend catering

Research Question

Do firms increase dividends
when shareholders are reallocating their portfolios?

I need periods when shareholders reconsider their portfolios

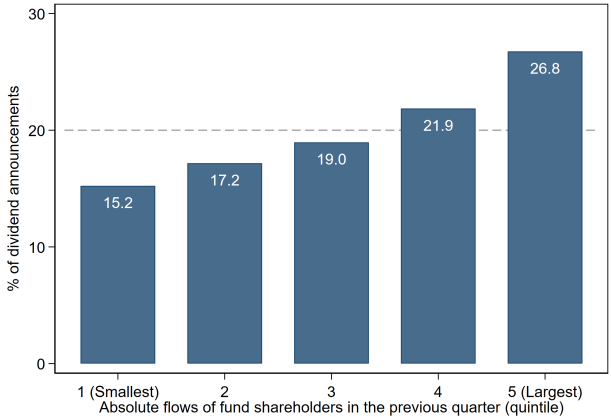
- Why not realized trading?
  - Often mechanical: funds trade in proportion to existing holdings (Lou, 2012)
  - May reflect firm-specific news that also drives payouts

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- Why not realized trading?
  - Often mechanical: funds trade in proportion to existing holdings (Lou, 2012)
  - May reflect firm-specific news that also drives payouts
- My approach: large mutual fund flows
  - Determined at the fund level, not by a particular holding
  - Large flows are **less likely** to be absorbed by proportional trading

    Large inflows or outflows can be very disruptive and may necessitate adjustments in portfolio management strategies. (Morningstar, 2017)

When do firms announce dividend increases?



- Announcements are **nearly twice** as frequent right after the largest flows.

Preview of results

1. Large fund flows lead to active portfolio rebalancing
   - Discretionary trading rises with **both large inflows and outflows**

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   - Morningstar Sustainability Rating introduction (2016): treated firms raise dividends
   - **Symmetric** across inflow and outflow shocks

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   - Stronger for firms whose shareholders prefer dividends: catering
4. Funds respond to dividend increases
   - Funds facing rating-induced flows trade toward treated firms that raise dividends

Contribution to the literature

1. **Dividend catering** (Baker and Wurgler, 2004; Li and Lie, 2006; Chetty and Saez, 2005; Grinstein and Michaely, 2005; Becker et al., 2011; Desai and Jin, 2011; Hanlon and Hoopes, 2014; Golubov et al., 2020)
   This paper: **When** catering occurs – when shareholders are reconsidering their portfolios.

2. **Real effects of mutual fund flows** Price pressure (Edmans et al., 2012; Khan et al., 2012; Lou and Wang, 2018) Fragility (Friberg et al., 2024; Massa et al., 2025)
   This paper: A **reallocation** channel, common to inflows and outflows.

3. **Flow-induced trading** (Coval and Stafford, 2007; Lou, 2012; Berger, 2023; Huang et al., 2023)
   This paper: Fund managers' response to flows varies with flow **magnitude**.

4. **Investor attention and corporate behavior** (Cohen and Frazzini, 2008; Hirshleifer et al., 2009; Kempf et al., 2017; Schmidt, 2019; Iliev et al., 2021; Hirshleifer and Sheng, 2022)
   This paper: Attention creates an incentive to **appeal to** shareholders, not only to be disciplined.

Outline

1. Data and sample
2. Fund flows and discretionary trading
3. Fund flows and dividend increases
4. Natural experiment: Morningstar Sustainability Rating
5. Fund responses to dividend increases

Data and sample

- **Data sources**
  - **Morningstar**: global open-end equity funds (TNA, flows, etc.)
  - **FactSet**: fund-level holdings
  - **Worldscope / Datastream**: firm financials and stock returns
- **Sample period**: 2000 – 2021
- **Funds**: active equity funds with TNA $\geq$ \$15 million and age $\geq$ 3 years; require flow data and lagged holdings.
- **Firms**: exclude financials (SIC 6000–6999) and utilities (SIC 4900–4999); exclude firms not covered by FactSet or not held by any eligible fund.
- Final sample: **14,593 funds** and **17,853 firms** across countries.

summary statistics

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Discretionary trading intensity

Discretionary Trading Intensity (DTI)

$$DTI_{f,q} = \sum_{s} \left(\left| Trading_{f,s,q} - ProRataTrading_{f,s,q} \right| \times Price_{s,q-1} \right)$$

Trading beyond what proportional scaling of flows would imply

Realized Trading

$$Trading_{f,s,q} = \Delta Holding_{f,s,q}$$

$Holding_{f,s,q}$: shares of stock s held by fund f at the end of quarter q

Pro-rata Trading example

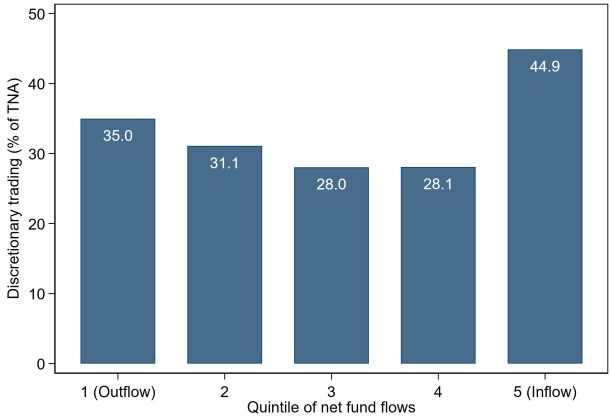
$$ProRataTrading_{f,s,q} = Holding_{f,s,q-1} \times Flow_{f,q}$$

where $Flow_{f,q} = \left(TNA_{f,q} - (1 + Ret_{f,q})\, TNA_{f,q-1} \right) / TNA_{f,q-1}$

Benchmark: funds allocate flows in proportion to existing positions

(Coval and Stafford, 2007; Lou, 2012)

Discretionary trading is U-shaped in net flows



- Discretionary trading depends on the **magnitude** of flows ⇒ use $|Flow|$.

Bigger flows, more active trading decisions

Dependent variable	$\ln(DTI)$			$\ln(DTI/TNA)$	
Sample	Full	$Flow > 0$	$Flow < 0$	$Flow > 0$	$Flow < 0$
	(1)	(2)	(3)	(4)	(5)
$\lvert Flow \rvert_t$	2.4896***	2.6741***	1.8270***	2.5959***	1.8172***
	(0.0319)	(0.0301)	(0.0476)	(0.0273)	(0.0471)
$\ln(TNA)_{t-1}$	0.8859***	0.8797***	0.9057***	-0.0974***	-0.0794***
	(0.0052)	(0.0067)	(0.0060)	(0.0059)	(0.0055)
$Return_{t-1}$	-0.1770***	-0.1199	-0.4825***	-0.0755	-0.4739***
	(0.0663)	(0.0953)	(0.0788)	(0.0941)	(0.0787)
Observations	500,195	168,931	329,748	168,931	329,748
Adjusted R-squared	0.7278	0.7246	0.7353	0.3887	0.3272
Fund FE	Y	Y	Y	Y	Y
Category × Year-Quarter FE	Y	Y	Y	Y	Y

- 1 s.d. ↑ in absolute flows ⇒ 32% ↑ in discretionary trading.

alternative measures portfolio overlap

Discretionary trading increases in both inflows and outflows

Dependent variable	$\ln(DTI)$			$\ln(DTI/TNA)$	
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- Not driven by larger funds trading larger dollar amounts (Columns 4–5).

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Measuring exposure to fund flows: reallocation vs. price

- Firm-level exposure to flow-induced reallocation:

$$\textit{Trading Pressure}_{s,q} = \frac{\sum_f \textit{Holding}_{f,s,q-1} \times |\textit{Flow}|_{f,q}}{\textit{Shares Outstanding}_{s,q-1}}$$

- Firm-level exposure to flow-induced price pressure (Coval and Stafford, 2007):

$$\textit{Price Pressure}_{s,q} = \frac{\sum_f \textit{Holding}_{f,s,q-1} \times \textit{Flow}_{f,q}}{\textit{Shares Outstanding}_{s,q-1}}$$

- Different predictions
  - Reallocation: **same** response to inflows and outflows.
  - Price: **opposite** responses to inflows and outflows.

- Both measures are standardized (mean 0, s.d. 1).

Specification: dividends and exposure to fund flows

- **Annual** panel, firm-year

$$\textit{Dividend}_{s,t} = \beta\, \textit{Trading Pressure}_{s,t-1} + \Gamma' \textit{Controls}_{s,t-1} + \alpha_s + \alpha_{c,i,t} + \varepsilon_{s,t}$$

- *Dividend*: dividends / beginning-of-year assets
- *Trading Pressure*: summed over the fiscal year
- *Controls*: firm characteristics (e.g., size, leverage, market-to-book, profitability)
- α_s: firm FE; $\alpha_{c,i,t}$: country × industry × year FE

Firms exposed to large fund flows increase dividends

Dependent variable	*Dividend* (1)	*Dividend* (2)	*Dividend* (3)	*Repurchase* (4)
Trading Pressure	0.0003**	0.0003**		
	(0.0001)	(0.0001)		
Price Pressure		0.0000		
		(0.0001)		
Trading Pressure$_{inflow}$			0.0002**	0.0001
			(0.0001)	(0.0001)
Trading Pressure$_{outflow}$			0.0002*	0.0006***
			(0.0001)	(0.0002)
Observations	147,110	147,110	147,110	147,110
Adjusted R-squared	0.7259	0.7259	0.7259	0.3893
Controls	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Country × Industry × Year FE	Y	Y	Y	Y

– 1 s.d. *Trading Pressure* ↑ ⇒ 1.5% dividend ↑ relative to the mean.

Dividends: symmetric. Repurchases: outflows only.

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Firm FE	Y	Y	Y	Y
Country × Industry × Year FE	Y	Y	Y	Y

– Dividends respond to reallocation; repurchases respond to price pressure (market timing).

Funds experiencing extreme flows drive the results

Dependent variable	*Dividend*	
	(1)	(2)
Trading Pressure^{extreme}	0.0003***	
	(0.0001)	
Trading Pressure^{normal}	-0.0001	
	(0.0002)	
Trading Pressure$^{extreme}_{inflow}$		0.0002***
		(0.0001)
Trading Pressure$^{extreme}_{outflow}$		0.0003***
		(0.0001)
Trading Pressure$^{normal}_{inflow}$		-0.0001
		(0.0001)
Trading Pressure$^{normal}_{outflow}$		-0.0001
		(0.0001)
Observations	147,110	147,110
Adjusted R-squared	0.7259	0.7260
Controls	Y	Y
Firm FE	Y	Y
Country × Industry × Year FE	Y	Y

- Extreme: flows above the 90th or below the 10th percentile in the quarter.

Symmetry holds within extreme flows only

Dependent variable	*Dividend*	
	(1)	(2)
Trading Pressure^{extreme}	0.0003***	
	(0.0001)	
Trading Pressure^{normal}	-0.0001	
	(0.0002)	
Trading Pressure$^{extreme}_{inflow}$		0.0002***
		(0.0001)
Trading Pressure$^{extreme}_{outflow}$		0.0003***
		(0.0001)
Trading Pressure$^{normal}_{inflow}$		-0.0001
		(0.0001)
Trading Pressure$^{normal}_{outflow}$		-0.0001
		(0.0001)
Observations	147,110	147,110
Adjusted R-squared	0.7259	0.7260
Controls	Y	Y
Firm FE	Y	Y
Country × Industry × Year FE	Y	Y

Specification: dividend announcement timing

- **Quarterly** panel, firm-quarter

$$1(\textit{Dividend Announcement})_{s,q} = \sum_{k=0}^{3} \beta_k \, \textit{Trading Pressure}_{s,q-k} + \alpha_{s,y} + \alpha_{c,i,q} + \varepsilon_{s,q}$$

- $1(\textit{Dividend Announcement})$: = 1 if the firm announces a dividend in quarter q
- *Trading Pressure*: measured in quarter q and the three quarters before
- $\alpha_{s,y}$: firm × year FE, so timing is compared **within the same firm-year**
- $\alpha_{c,i,q}$: country × industry × year-quarter FE

Firms choose the timing of dividend increases

Dependent variable	$1(\textit{Dividend Announcement})_q$			
Sample	Full	Dividend Increase	No Change	Dividend Decrease
	(1)	(2)	(3)	(4)
Trading Pressure$_q$	0.0010	0.0024	-0.0001	-0.0006
	(0.0008)	(0.0016)	(0.0002)	(0.0027)
Trading Pressure$_{q-1}$	0.0023***	0.0035**	0.0001	0.0024
	(0.0008)	(0.0016)	(0.0002)	(0.0025)
Trading Pressure$_{q-2}$	-0.0010	0.0003	-0.0002	-0.0037
	(0.0008)	(0.0017)	(0.0002)	(0.0026)
Trading Pressure$_{q-3}$	-0.0003	-0.0010	-0.0001	-0.0022
	(0.0009)	(0.0017)	(0.0002)	(0.0026)
Observations	716,184	309,848	189,302	174,180
Adjusted R-squared	0.5302	0.5504	0.5706	0.4643
Firm × Year FE	Y	Y	Y	Y
Country × Industry × Year-Quarter FE	Y	Y	Y	Y

- Only the **preceding quarter** predicts announcements, driven by dividend **increases**.

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- March 2016: Morningstar assigns 1–5 “Globes” to 20,000+ funds ($8 trillion).
  - Based on pre-existing holdings; no change in fundamentals of funds or firms.
  - Five-globe funds: inflows; one-globe funds: outflows (Hartzmark and Sussman, 2019) validation
  - Inflow and outflow shocks from a **single event** ⇒ test symmetry.

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- **Treated**: top quartile of 2015 ownership by five-globe or one-globe funds.

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  - Inflow and outflow shocks from a **single event** ⇒ test symmetry.
- **Treated**: top quartile of 2015 ownership by five-globe or one-globe funds.
- If the flow shock induces firms to increase dividends, treated firms should raise dividends following the event.

Treated firms raise dividends after rating publication

Dependent variable	*Dividend*		
	(1)	(2)	(3)
Treat × *Post*	0.0024***		
	(0.0006)		
Treat × Pre_{2015}		0.0005	
		(0.0007)	
Treat × $Post_{2016}$		0.0023***	
		(0.0009)	
Treat × $Post_{2017}$		0.0028***	
		(0.0009)	
Treat × $Post_{2018}$		0.0028***	
		(0.0009)	
$Treat_{FiveGlobes}$ × *Post*			0.0016**
			(0.0007)
$Treat_{OneGlobe}$ × *Post*			0.0020***
			(0.0006)
Observations	38,133	38,133	38,133
Adjusted R-squared	0.8173	0.8172	0.8173
Pre-event controls × Year	Y	Y	Y
Firm FE	Y	Y	Y
Country × Industry × Year FE	Y	Y	Y

- +0.24 pp of assets (≈ 12% of the mean); no pre-trend; present from FY2016 onward.

Dividend increases are symmetric across inflow and outflow shocks

Dependent variable	*Dividend*		
	(1)	(2)	(3)
Treat × *Post*	0.0024***		
	(0.0006)		
Treat × Pre_{2015}		0.0005	
		(0.0007)	
Treat × $Post_{2016}$		0.0023***	
		(0.0009)	
Treat × $Post_{2017}$		0.0028***	
		(0.0009)	
Treat × $Post_{2018}$		0.0028***	
		(0.0009)	
$Treat_{FiveGlobes}$ × *Post*			0.0016**
			(0.0007)
$Treat_{OneGlobe}$ × *Post*			0.0020***
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Adjusted R-squared	0.8173	0.8172	0.8173
Pre-event controls × Year	Y	Y	Y
Firm FE	Y	Y	Y
Country × Industry × Year FE	Y	Y	Y

– Opposite price effects, same dividend response ⇒ firms respond to reassessment, not prices.

Is it catering?

1. Catering channel (Baker and Wurgler, 2004; Li and Lie, 2006; Golubov et al., 2020)
   - Firms raise dividends to cater to shareholders' preferences while they reallocate.
   - Prediction: stronger when shareholders **prefer dividends**.

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2. Alternative: Governance channel (Easterbrook, 1984; Jensen, 1986)
   - Reallocation intensifies monitoring and the threat of exit.
   - Dividends commit free cash flow away from managers.
   - Prediction: stronger with **more free cash flow** and **fewer investment opportunities**.

Evidence is consistent with the catering channel

Dependent variable	*Dividend*		
	(1)	(2)	(3)
Treat × *Post*	0.0012*	0.0016**	0.0012*
	(0.0007)	(0.0008)	(0.0007)
× $D(High\ Own_{Income\ Funds})$	0.0033**		
	(0.0013)		
× $D(High\ Own_{Tax\ Cut})$		0.0025**	
		(0.0012)	
× $D(High\ Own_{Older\ Populations})$			0.0022*
			(0.0011)
Observations	35,595	35,212	35,236
Adjusted R-squared	0.8199	0.8212	0.8220
Pre-event controls × Year × Group	Y	Y	Y
Firm FE × Group	Y	Y	Y
Country × Industry × Year FE × Group	Y	Y	Y

- Firms with greater ownership by shareholders who prefer dividends show stronger responses.

No support for the governance channel

Dependent variable	*Dividend*					
	Free cash flow			Investment opportunities		
	(1)	(2)	(3)	(4)	(5)	(6)
Treat × *Post*	0.0031***	0.0023**	0.0022***	0.0029***	0.0026***	0.0031***
	(0.0009)	(0.0009)	(0.0006)	(0.0010)	(0.0009)	(0.0008)
×*D*(*Low Debt*)	-0.0005					
	(0.0012)					
×*D*(*High Cash*)		0.0000				
		(0.0012)				
×*D*(*High FCF*)			0.0010			
			(0.0013)			
×*D*(*Low M/B*)				-0.0013		
				(0.0013)		
×*D*(*Low Sales Growth*)					-0.0005	
					(0.0012)	
×*D*(*Low Investment*)						-0.0014
						(0.0012)
Observations	32,630	32,438	32,405	32,560	32,592	32,559
Adjusted R-squared	0.8138	0.8150	0.8163	0.8151	0.8117	0.8152
Pre-event controls × Year × Group	Y	Y	Y	Y	Y	Y
Firm FE × Group	Y	Y	Y	Y	Y	Y
Country × Industry × Year FE × Group	Y	Y	Y	Y	Y	Y

- None significant; investment-opportunity proxies have the **opposite** sign.

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Does catering work? Fund trading after dividend increases

- Fund-firm-quarter panel of five-globe and one-globe funds

$$\begin{aligned} Trading_{f,s,q} = {} & \beta_1 Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0) \\ & + \beta_2 Treat_s \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0) \\ & + \Gamma'(\bar{X}_{s,pre} \times \tau_q) + \alpha_s + \alpha_{f,q} + \alpha_{Treat \times Post} + \varepsilon_{f,s,q} \end{aligned}$$

- *Trading*: fund's quarterly trading in the stock, scaled by lagged shares outstanding
- $I(\Delta Dividend_{2015-2016} > 0)$: firm raised its dividend from FY2015 to FY2016
- $Post_{>2016}$: quarters after the end of the firm's FY2016
- β_2: trading toward **treated** firms that raise dividends, relative to control firms

Funds trade toward treated firms that raise dividends

Dependent variable	*Trading*		
Fund sample	Five and One (1)	Five (2)	One (3)
$Treat \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$	0.0030*** (0.0010)		
$Treat_{FiveGlobes} \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$		0.0056*** (0.0017)	
$Treat_{OneGlobe} \times Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$			0.0025** (0.0012)
Observations	1,256,253	546,946	708,567
Adjusted R-squared	0.0770	0.0760	0.0853
$Post_{>2016} \times I(\Delta Dividend_{2015-2016} > 0)$	Y	Y	Y
Treat × *Post* FE	Y	Y	Y
Pre-event controls × Year-Quarter	Y	Y	Y
Firm FE	Y	Y	Y
Fund × Year-Quarter FE	Y	Y	Y

- Five-globe funds (new capital): dividend increases attract capital.
- One-globe funds (redemptions): dividend increases retain capital.

Conclusion

- Firms increase dividends when shareholders reallocate their portfolios.
- Large flows push fund managers into discretionary trading.
- Firms exposed to large flows increase dividends.
  - Symmetric across inflows and outflows; announced in the following quarter.
  - Causal evidence from the Morningstar Sustainability Rating.
  - Stronger for firms whose shareholders prefer dividends: catering.
- Funds trade toward firms that raise dividends.
- **Implication**: firms choose not only **how much** to pay, but also **when**.

Thank you!

Summary statistics

Panel A: Fund Sample						
	Obs	Mean	STD	P25	Median	P75
DTI ($bil)	500,195	0.2009	0.4788	0.0112	0.0419	0.1553
ln (*DTI*)	500,195	-3.2365	1.9872	-4.4959	-3.1730	-1.8622
Flow	500,195	-0.0042	0.1274	-0.0516	-0.0188	0.0179
|*Flow*|	500,195	0.0764	0.1114	0.0186	0.0406	0.0840
TNA ($bil)	500,195	0.7603	1.7422	0.0619	0.1824	0.6027
Return	500,195	0.0267	0.1011	-0.0193	0.0336	0.0835
Age (years)	500,195	15.1691	10.3654	8	13	19.75
Panel B: Firm Sample						
	Obs	Mean	STD	P25	Median	P75
Dividend	147,110	0.0196	0.0318	0	0.0087	0.0237
Repurchase	147,110	0.0087	0.0263	0	0	0.0015
Trading Pressure	147,110	0.0166	0.0200	0.0016	0.0082	0.0254
Price Pressure	147,110	-0.0015	0.0113	-0.0046	-0.0002	0.0010
Assets ($bil)	147,110	3.2700	8.6603	0.2107	0.6090	2.0604
Debt	147,110	0.2217	0.2021	0.0413	0.1859	0.3434
M/*B*	147,110	2.5677	3.0200	0.9229	1.6038	2.9316
ROE	147,110	0.0486	0.2784	0.0186	0.0836	0.1576
IO	147,110	0.1670	0.1599	0.0373	0.1112	0.2575
Cash	147,110	0.2094	0.2104	0.0646	0.1437	0.2798
Intangible	147,110	0.1321	0.2023	0.0057	0.0338	0.1758
Investment	147,110	0.0814	0.0890	0.0248	0.0542	0.1038
SGA/*Sale*	147,110	0.3498	0.7917	0.1025	0.1897	0.3364

back

Example: Pro-rata trading

- Initially:

Stock	Holding	Weight
A	100	50%
B	100	50%

- Suppose that the fund receives a 10% flow.

Stock	Holding	Weight
A	$100 \times (1 + 10\%) = 110$	50%
B	$100 \times (1 + 10\%) = 110$	50%

- All positions are scaled proportionally, thereby preserving the portfolio composition.

back

Alternative measures of discretionary trading

Sample	Full Sample (1)	*Flow* > 0 (2)	*Flow* < 0 (3)
Panel A: *DTI*/*TNA*			
|*Flow*|	0.7839***	1.0337***	0.1434***
	(0.0111)	(0.0126)	(0.0105)
Observations	500,201	168,931	329,754
Adjusted R-squared	0.3888	0.4790	0.3530
Panel B: *DTI*/*SharesOutstanding*			
|*Flow*|	1.3077***	1.6759***	0.4349**
	(0.1347)	(0.1716)	(0.1744)
Observations	500,186	168,924	329,746
Adjusted R-squared	0.2152	0.2572	0.2089
Panel C: *High DTI Ratio*			
|*Flow*|	0.3622***	0.3937***	0.0605***
	(0.0120)	(0.0137)	(0.0136)
Observations	414,726	138,800	274,081
Adjusted R-squared	0.1913	0.1995	0.2248
Controls	Y	Y	Y
Fund FE	Y	Y	Y
Category × Year-Quarter FE	Y	Y	Y

Panel C: *High DTI Ratio* = 1 if *DTI* / total trading is in the top 20% of the sample. back

Larger flows change portfolio weights

$$Overlap_{f,q} = 1 - \frac{1}{2} \sum_i \left| w_{i,f,q} - w_{i,f,q-1} \right|$$

Dependent variable	*Overlap*		
Sample	Full	*Flow* > 0	*Flow* < 0
	(1)	(2)	(3)
$\lvert Flow \rvert_t$	−0.0728***	−0.0578***	−0.1041***
	(0.0022)	(0.0028)	(0.0037)
$\ln(TNA)_{t-1}$	0.0086***	0.0097***	0.0090***
	(0.0005)	(0.0006)	(0.0006)
$Return_{t-1}$	0.0674***	0.0271***	0.0791***
	(0.0057)	(0.0099)	(0.0073)
Observations	500,201	168,931	329,754
Adjusted R-squared	0.2910	0.3103	0.2902
Fund FE	Y	Y	Y
Category × Year-Quarter FE	Y	Y	Y

Addresses the concern that *DTI* captures trades that merely restore existing weights. back

Placebo: passive fund flows

Dependent variable	*Dividend*	
	(1)	(2)
Trading Pressure^{active}	0.0003**	
	(0.0001)	
Trading Pressure^{passive}	0.0004***	
	(0.0002)	
Trading Pressure$^{active}_{inflow}$		0.0002**
		(0.0001)
Trading Pressure$^{active}_{outflow}$		0.0002**
		(0.0001)
Trading Pressure$^{passive}_{inflow}$		0.0000
		(0.0001)
Trading Pressure$^{passive}_{outflow}$		0.0005***
		(0.0001)
Observations	152,135	152,135
Adjusted R-squared	0.7246	0.7246
Controls	Y	Y
Firm FE	Y	Y
Country × Industry × Year FE	Y	Y

- Active funds: symmetric. Passive funds: outflows only, like the repurchase pattern.

back

Validation: globe ratings move flows and discretionary trading

Dependent variable	*Flow*		*High DTI Ratio*	
Sample	Full (1)	Rated (2)	Full (3)	Rated (4)
Five Globes × $Post_{>2016Q1}$	0.0111***	0.0088**	0.0236**	0.0235**
	(0.0036)	(0.0037)	(0.0111)	(0.0112)
One Globe × $Post_{>2016Q1}$	-0.0073*	-0.0097**	0.0230**	0.0228**
	(0.0041)	(0.0042)	(0.0111)	(0.0111)
Observations	65,573	58,845	57,881	52,323
Adjusted R-squared	0.2946	0.2920	0.2544	0.2565
Controls	Y	Y	Y	Y
Fund FE	Y	Y	Y	Y
Category × Year-Quarter FE	Y	Y	Y	Y

- Each fund is assigned its earliest available rating (2018 methodology revision).
- Both groups become about 2.3 pp more likely to have a high *DTI* ratio.

back

Dynamics by inflow and outflow shocks

Dependent variable	*Dividend*	
	(1)	(2)
$Treat_{FiveGlobes} \times Pre_{2015}$	0.0009	0.0006
	(0.0007)	(0.0008)
$Treat_{FiveGlobes} \times Post_{2016}$	0.0014	0.0011
	(0.0009)	(0.0010)
$Treat_{FiveGlobes} \times Post_{2017}$	0.0029***	0.0027**
	(0.0009)	(0.0011)
$Treat_{FiveGlobes} \times Post_{2018}$	0.0018*	0.0022*
	(0.0010)	(0.0012)
$Treat_{OneGlobe} \times Pre_{2015}$	-0.0002	-0.0001
	(0.0006)	(0.0006)
$Treat_{OneGlobe} \times Post_{2016}$	0.0017**	0.0020***
	(0.0007)	(0.0008)
$Treat_{OneGlobe} \times Post_{2017}$	0.0012	0.0016**
	(0.0008)	(0.0008)
$Treat_{OneGlobe} \times Post_{2018}$	0.0007	0.0021**
	(0.0008)	(0.0008)
Observations	38,133	38,133
Adjusted R-squared	0.8162	0.8173
Pre-event Controls × Year	N	Y
Firm FE	Y	Y
Country × Industry × Year FE	Y	Y

back

Dividend increases and initiations

Dependent variable	*Increase*	*Increase*	*Initiation*	*Initiation*
Sample	Payers (1)	Payers (2)	Non-Payers (3)	Non-Payers (4)
Treat × *Post*	0.0126		0.0168	
	(0.0167)		(0.0122)	
Treat × Pre_{2015}		-0.0013		0.0273
		(0.0257)		(0.0183)
Treat × $Post_{2016}$		0.0505*		0.0313
		(0.0269)		(0.0208)
Treat × $Post_{2017}$		0.0030		0.0516**
		(0.0270)		(0.0207)
Treat × $Post_{2018}$		-0.0225		0.0240
		(0.0267)		(0.0230)
Observations	26,413	26,413	10,232	10,232
Adjusted R-squared	0.1686	0.1688	-0.0063	-0.0063
Pre-event controls × Year	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Country × Industry × Year FE	Y	Y	Y	Y

– Payers: +5.1 pp more likely to raise dividends in 2016. Non-payers: +5.2 pp more likely to initiate in 2017.

back

US vs. non-US firms

	US	Non-US
Dividends on extreme-flow trading pressure (annual)		
Trading Pressure^{extreme}	0.0003**	0.0002**
	(0.0001)	(0.0001)
Observations	38,311	108,799
Difference-in-differences		
Treat × *Post*	0.0061***	0.0019***
	(0.0020)	(0.0006)
Observations	6,601	31,532
Controls, Firm FE, Country × Industry × Year FE	Y	Y

- The extreme-flow effect and the DiD result hold in both US and non-US firms.

back to extreme flows back to DiD